

Internet of Things (IoT)

Connecting everyday objects to the internet

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The Internet of Things connects physical devices — from thermostats to industrial machinery — to the internet, enabling them to collect and exchange data automatically.

● DEFINITION

A network of physical objects embedded with sensors, software, and connectivity that enables them to collect and exchange data over the internet.

● WHY IT MATTERS

IoT enables real-time monitoring and automation across homes, cities, and industries, transforming previously 'dumb' objects into data-generating, connected devices.

● WORKING PRINCIPLE



IoT devices use embedded sensors to collect data about their environment or status, transmitting it over networks to cloud or local systems that can analyze the data and trigger automated responses.

● QUICK FACTS

- ▶ There are estimated to be tens of billions of connected IoT devices worldwide
- ▶ The term 'Internet of Things' was coined by Kevin Ashton in 1999

● KEY TERMS

- Sensor
- Edge computing
- Connectivity protocol
- Smart device

● CORE CONCEPTS

- ▶ Sensor networks
- ▶ Device connectivity protocols
- ▶ Edge computing
- ▶ Data collection and analytics

● APPLICATIONS

- ▶ Smart home automation
- ▶ Industrial equipment monitoring
- ▶ Smart city infrastructure
- ▶ Connected healthcare wearables

● ADVANTAGES

- ▶ Enables real-time monitoring and automation
- ▶ Generates valuable data for optimization
- ▶ Improves efficiency across homes and industries

CAREER OPPORTUNITIES

IoT engineer, Embedded systems developer, IoT security specialist, Smart systems architect

RESEARCH AREAS

IoT security, Edge computing for IoT, Low-power IoT communication protocols

● CURRENT TRENDS

Rapid growth in industrial IoT applications, using connected sensors to predict equipment failures before they happen.

● REAL-LIFE EXAMPLES

- ▶ Smart thermostats learning household patterns
- ▶ Fitness trackers monitoring heart rate
- ▶ Smart streetlights adjusting to traffic

● LIMITATIONS

- ▶ Security vulnerabilities in connected devices are a major concern
- ▶ Device interoperability standards vary
- ▶ Requires reliable connectivity infrastructure

INDUSTRY APPLICATIONS

Smart homes, Manufacturing, Agriculture, Healthcare

FUTURE SCOPE

Continued growth in connected devices, combined with 5G networks and edge computing, is expected to expand IoT's real-time capabilities significantly.

● SUMMARY

The Internet of Things connects everyday objects to the internet, enabling real-time data collection and automation across homes, cities, and industry.